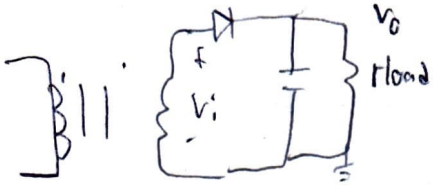
Rylan Thornhing

1.



31.8 p dop → 15.9 mag

120 rms = $120\sqrt{2}$ mag

$V_f = 1.1V$

1. $\frac{5}{2500} = 2mA$ $\boxed{I_{Load} = 2mA}$

2. $\frac{120\sqrt{2}}{15.9} = 10.67$

$\boxed{10.67:1}$

3. $V_r(pp) = \frac{1}{fRC}(V_p)$

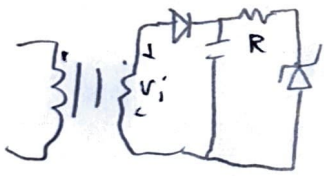
If, $V_p = 15.9 - 1.1 = 14.8$ If $V_p = 5$

$2 = \frac{14.8}{60(2500)C}$ $2 = \frac{5}{60(2500)C}$

$C = 49.33\mu F$ $\boxed{C = 16.67\mu F}$

I'm more confident in this answer

2.



dynamic zener res: 17Ω

test data point: 28mA

$15.9 - 1.1 = 14.8$

$\frac{14.8 - 5}{R} = .002 + .0035$

$\frac{9.8}{R} = .0055$

$\boxed{R = 1781.82\Omega}$

Questions

1. I can use an oscilliscope to make sure my rectifier is cutting off negative (really <-.7V and slight ripple) voltages.

2. If the difference between the peak and lowest point of the wave form after the first period is less than the V_r specified, it is being met.

3. If I don't want to measure the load current, I could always measure load voltage and divide by resistance to check.

4. I would measure the V_r before and after adding the Zener diode and compare.

5. I would measure load regulation by using different resistors and seeing what happens to the voltage across the resistor. This will also tell me current after I take the measured voltage and divide it by R_{load}.

6. I would look up the datasheet for my load resistor or component and find the max load current for it. This will be my max load current for the whole psu